WanderLog — Final ER Diagram

Chen notation · cardinalities read across the relationship (e.g. 1 user — N posts)

Entity
Relationship
Attribute (underlined = key)
Double line = total participation (every instance takes part)
Single line = partial participation (optional)
ISA = subtype (apex points to the supertype)

MODERATORS
user_id
moderator_since
ISA
REVIEWS
SUBMITS
USERS
user_id
username
email
password_hash
display_name
bio
account_status
created_at
updated_at
FOLLOWS
follower
following
followed_at
CREATES
POSTS
post_id
title
content
location
created_at
updated_at
LIKES_POST
liked_at
HAS_COMMENT
ABOUT_POST
exactly one
WRITES_COMMENT
COMMENTS
comment_id
created_at
content
updated_at
LIKES_COMMENT
liked_at
ABOUT_COMMENT
REPORTS
report_id
reason
status
created_at
WRITES_REPLY
REPLIES_TO
ABOUT_REPLY
REPLIES
reply_id
content
created_at
updated_at

Implementation notes (relational mapping)

- Likes are M:N → junction tables post_likes / comment_likes (user_id, post_id | comment_id, liked_at), composite PK.
- FOLLOWS is a self-referencing M:N; CHECK follower ≠ following.
- MODERATORS is a subtype of USERS (user_id is PK and FK).
- Each report targets exactly one of post / comment / reply (3 nullable FKs + CHECK exactly one is NOT NULL).
- REVIEWS is optional per report: reviewed_by is NULL until a moderator picks it up.
- username and email are UNIQUE.
- Only posts and comments can be liked (replies cannot).
- liked_at is stored on the like relationships (post_likes / comment_likes).
- Double lines (total participation) = NOT NULL foreign keys: posts.user_id, comments.user_id and post_id, replies.user_id and comment_id, reports.user_id.